

User's Manual

BIOLAB MANAGER

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Overview:

The BioLab Manager Application is a text-based console application (Command-Line Interface) designed to manage users, clinical tests, and operations within a clinical laboratory system. The application supports Administrators, Physicians, Laboratory Technicians, and Patients, offering specialized, role-specific text menus after a secure login.

Starting the Application

1. Launching

To start the application:

- Run the main application file (e.g., Main.java).
- The application will print the main menu banner in your console/terminal.

2. Login

- Select option 1. Enter system by typing 1 and pressing Enter.
- Type your assigned Username and press Enter.
- Type your Password and press Enter.
- The system will securely authenticate your credentials using JPA and automatically detect your assigned role (ADMIN, PHYSICIAN, LAB_TECHNICIAN, or PATIENT).
- Successful login clears the screen and prints a new text menu specific to your role.

Note: The system strictly enforces Role-Based Access Control (RBAC). You will only see the menu options you are authorized to use.

Role Based-Menus

1. Administrator user manual

The Administrator module is designed to allow users to manage the system's database, oversee user accounts, and handle data interoperability (XML/HTML).

Patients Management

- ◇ Access the sub-menu to manage the patient census.
- ◇ Options: Add a new patient, view the full list, search by ID, update demographic data, or delete records.

• Physicians Management

- ◇ Management of the authorized medical staff.
- ◇ Options: Register new doctors, edit specialties and workplaces, or delete medical profiles.

• Laboratory Orders Oversight

- ◇ View and control all laboratory orders generated in the system, regardless of the prescribing doctor or patient.

• Tests Catalog Configuration

- ◇ Manage the catalog of available tests (e.g., Glucose, Cholesterol). Allows defining the name and description of each analysis.

• Reports & Analytics (OrderTest)

- ◇ Allows the admin to view the detailed relationship of test results and cross-reference data with healthy ranges.

• Export Database to XML

- ◇ Creates an XML backup of the laboratory database.
- ◇ Steps: Type the number for "Export XML". A confirmation message will be printed on the console once the export.xml file is successfully generated.

• Generate HTML Report

- ◇ Converts the exported XML data into a readable HTML format using XSLT.
- ◇ Steps: Type the number for "Generate HTML". The system will process the biolab.xml file and generate a report.html file in your project folder.

• Import Database from XML

- ◇ Loads an existing XML file into the database. Use with caution, as this updates existing database records.

2. Physician User Manual

The Physician Menu is designed for medical professionals to manage their patients and prescribe clinical tests.

- **My Patients Overview**

- ◇ Type the option for "Patients" to view a list of all patients assigned to your medical profile.

- **Prescribe Laboratory Order**

- ◇ Generate a new test order for a patient.
- ◇ Steps:
 1. Access the "Orders" menu.
 2. Follow the console prompts to enter the Patient ID.
 3. Type the ID of the required test (e.g., Glucose, Hemoglobin).
 4. Confirm the input. The order will be saved to the database as "Pending".

- **View Orders / Analytics**

- ◇ Track the status of the orders you have prescribed, and review completed clinical results directly in the console.

3. Laboratory Technician User Manual

The Lab Technician Menu focuses entirely on the operational side of the laboratory, managing pending orders and entering medical data.

- **Manage Test Catalog**

- ◇ Add new tests to the system or update Reference Ranges (healthy minimum and maximum values).

- **View Pending Laboratory Orders**

- ◇ Select "Manage Laboratory Orders" to print a list of all orders waiting to be processed.

- **Enter Test Results**

- ◇ Input the numeric results obtained from the physical lab machines.
- ◇ Steps:
 1. Select "Enter / Edit Test Results".
 2. Type the Order ID and Test ID when prompted.
 3. Input the numeric Result Value.
 4. Press Enter. The system will automatically compare the value against the Reference Range, assign a Status (e.g., NORMAL, ABNORMAL), and print a success message.

4. Patient user Manual

The Patient Menu is a strict read-only interface designed for users to safely monitor their personal health records.

- **View my results**

- ◇ View a history of all completed laboratory tests.
- ◇ Steps: Type 2 for "View my results". The console will print your records, showing the Order ID, Test ID, Value, and Status.

- **View my orders**

- ◇ Type 1 to check if your physician has prescribed new tests that are pending.

Note: If any test status appears as abnormal, please contact your prescribing physician. Patients cannot modify any medical data.

- **Account Security**

- ◇ Option to log out securely to protect the privacy of sensitive clinical data.

- **Data Integrity Summary**

To ensure medical system integrity, please note the following application rules:

- ◇ Orders can only be prescribed by Physicians.
- ◇ Results can only be inputted by Laboratory Technicians.
- ◇ Global Database Reports are the exclusive responsibility of the Administrator.

General Actions (All roles)

- Logout / Return
 - ◇ Logs out the current user securely and exits the current role menu.
 - ◇ Steps: Type 0 (or the designated exit number) for "Logout". You will be returned to the initial BioLab Manager main menu.

Troubleshooting

Issue	Solution
Invalid credentials	Ensure your username and password are typed correctly. The console will print "Invalid credentials" if incorrect.
"Invalid option" error	Ensure you are typing the exact number shown in the menu (e.g., typing "1" instead of the word "One").
"User not linked" error	Your login account is valid, but the Administrator has not linked your User ID to a physical Patient ID or Physician ID in the database. Contact Support.
XML Export fails	Ensure the application has write/read permissions in the folder where it is running.